



HCT: The HDL Complexity Tool

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HCT stands for HDL Complexity Tool, where HDL stands for Hardware Description Language. HCT provides scores that represent the complexity of modules present in integrated circuit (IC) designs. It is written in Perl and released under the GPLv3 and LGPLv3 license. It employs McCabe Cyclomatic Complexity that uses the control flow graph of the program source code to determine the complexity.

There are various factors for measuring the complexity of HDL models, such as size, nesting, modularity and timing. The measured metrics can help designers in refactoring their code, and also help managers to plan project schedules, and allocate resources, accordingly. You can run the tool from the Linux terminal for Verilog, VHDL and CDL (computer design language) files.

HCT can be installed on Fedora using the command:

```
$ sudo yum install hct
```

After installation, consider the example project of uart2spi written in Verilog, which is included in this month's EFY DVD. It implements a simple core for a UART interface and an internal SPI bus. The uart2spi folder contains rtl/spi under the file directory in your PC: /home/guest/uart2spi/trunk/rtl/spi

Run the HCT tool on the rtl/spi Verilog source as follows:

```
$ hct rtl/spi
```

We get the output:

```
Directory: /home/guest/uart2spi/trunk/rtl/spi
verilog, 4 file(s)
```

FILENAME	MODULE	IO	NET	MCCABE	TIME
spi_ctl.v		20	1	1	0.1724
	spi_ctl	20	1	1	
spi_core.v		0	0	1	0.0076
	spi_core	0	0	1	
spi_cfg.v		0	0	1	0.0076
	spi_cfg	0	0	1	
spi_if.v		15	3	1	0.0994
	spi_if	15	3	1	

The output includes various attributes that are described below:

1. **FILENAME** is the file that is being parsed. The parser uses the file name extension to recognize the programming language.

2. **MODULE** refers to the specific module present in the file. A file can contain many modules.

3. **IO** refers to the input/output registers used in the module.

4. **NET** includes the network entities declared in the given module. For Verilog, it can be 'wire', 'tri', 'supply0', etc.

5. **MCCABE** provides the McCabe Cyclomatic Complexity of the module or file.

6. **TIME** refers to the time taken to process the file.

A specific metric can be excluded from the output using the "-output-exclude=LIST" option. For example, type the following command on a Linux terminal:

```
$ hct --output-exclude=TIME rtl/spi
```

The output will be:

```
Directory: /home/guest/uart2spi/trunk/rtl/spi
verilog, 4 file(s)
```

FILENAME	MODULE	IO	NET	MCCABE
spi_ctl.v		20	1	1
	spi_ctl	20	1	1
spi_core.v		0	0	1
	spi_core	0	0	1
spi_cfg.v		0	0	1
	spi_cfg	0	0	1
spi_if.v		15	3	1
	spi_if	15	3	1

If you want only the scores to be listed, you can remove the **MODULE** listing with the "-output-no-modules" option:

```
$ hct --output-no-modules rtl/spi
```

```
Directory: /home/guest/uart2spi/trunk/rtl/spi
verilog, 4 file(s)
```

FILENAME	IO	NET	MCCABE	TIME
spi_ctl.v	20	1	1	0.16803
spi_core.v	0	0	1	0.007434
spi_cfg.v	0	0	1	0.00755
spi_if.v	15	3	1	0.097721

The tool can be run on individual files, or recursively on subdirectories with the "-R" option. The output of the entire